DA2402 Data Curation and Visualisation Project

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November 3, 2026

Statement on the Use of AI Tools

This project was carried out with the course material as the primary reference for the statistical concepts, data cleaning methods, and visualization principles applied, including the choice of outlier detection methods, missingness tests, imputation strategy, and visual design choices. AI assistants were used in a supporting role alongside this, mainly to speed up the mechanical and repetitive parts of the work, generate boilerplate code for data plotting, and help with LaTeX formatting.

Task	**Nature of AI assistance**
Data cleaning notebook	Used to draft and debug the Python code for header reshaping, data type correction, outlier detection (IQR, modified Z-score, Grubbs' test, DBSCAN, Local Outlier Factor) and correlation guided KNN imputation. The choice of which tests to run and how to justify each decision was made by the team, based on concepts covered in the course.
Data visualization and plotting	Used to assist in writing Python code (using Matplotlib/Seaborn) for exploratory charts and diagnostic figures, as well as refining plot aesthetics, axis formatting, color schemes, and layout parameters. The choice of chart types, channel encodings, and visual interpretations was performed entirely by the team.
Report writing and LaTeX formatting	Used to convert cleaning notebook outputs into LaTeX sections, tables and the data dictionary, and to fix formatting and compilation issues such as package errors.
Debugging and troubleshooting	Used to identify and fix code errors encountered while running the cleaning and visualization pipeline, such as package import issues and parameter tuning for the outlier detection methods.

All outputs produced with AI assistance were reviewed, verified against the actual notebook outputs, and edited by the team before being included in this report. No analysis, result or conclusion in this project was generated without being checked by the team members against the underlying data.